



Event Factuality in Italian: Annotation of News Stories from the Ita-TimeBank

ANNE-LYSE MINARD, ALESSANDRO MARCHETTI, MANUELA SPERANZA

Extension of the Ita-TimeBank with event factuality annotation on top of TimeML annotation.

CORPUS

- A subsection of Ita-TimeBank [1], first released as training data for the EVENTI Task at EVALITA (www.evalita.it/2014/tasks/eventi)
- Size: 170 news stories (65,455 tokens)
- Factuality annotation performed on top of Ita-TimeBank annotations:
 - annotation layers: time expression, event and temporal relation
 - annotation schema: It-TimeML [1], an adaptation of TimeML to Italian

Statistics about the sub-corpus of Ita-TimeBank

PoS	# events	classes	# events
verb	6,300	OCCURRENCE	5,292
noun	3,526	STATE	2,352
adjective	352	I_ACTION	900
preposition	27	I_STATE	864
		REPORTING	439
		ASPECTUAL	258
		PERCEPTION	100

EXAMPLES

Andrew Smith ha **rassegnato** ieri le dimissioni nonostante i **tentativi** del premier Tony Blair di **convincerlo** a rimanere.

[Andrew Smith resigned despite Tony Blair's attempts to convince him to stay.]

Non ha **nominato** un amministratore delegato per Mumbai.

[He did not appoint a CEO for Mumbai.]

REPRESENTATION OF EVENT FACTUALITY

Event factuality is represented on three main axes (based on the framework proposed by van Son et al., 2014 [2]):

- Certainty:** defines as how certain the source is about an event, with the following three values: *certain*, *possible*, *probable*.
- Time:** specifies the time an event took place or will take place. Its values are *non future* (for present and past events), *future* (for events that will take place), and *underspecified* (used for general events and when the time of an event cannot be determined).
- Polarity:** captures if an event is affirmed or negated and, consequently, it can be either *positive* or *negative*.

The attribute *special cases* is used in order to mark hypothetical events and general statements (possible values are: *conditional construction*, *general statement* and *none*).

Combining the three attributes certainty, time and polarity, and taking into account the special cases, we can determine whether the term considered refers to a factual, a counterfactual or a non factual event.

- FACTUAL:** *certain* + *positive* + *non future*
- COUNTERFACTUAL:** *certain* + *negative* + *non future/future*
- NON FACTUAL (uncertain):** *possible/probable* + *positive* + *non future/future*
- NON FACTUAL (future):** *certain* + *future* + *positive*
- NON FACTUAL (conditional/generic):** *conditional construction* / *general statement*

QUANTITATIVE DATA

	event classes					news topics				
	I_ACTION	REPORTING	PERCEPTION	OCCURRENCE	ASPECTUAL	Trento	Sport	Economy	Culture	News
# events	900	439	100	5,292	258	3,084	886	735	684	1,600
Factual (%)	65.2	84.5	66.0	69.0	65.5	68.2	71.1	66.4	62.9	74.6
Counterfactual (%)	3.8	2.7	8.0	3.8	1.6	4.5	4.4	1.4	2.5	3.5
Future - certain (%)	9.0	2.5	6.0	10.9	21.3	9.5	14.0	16.9	16.5	4.8
Future - uncertain (%)	14.2	6.6	12.0	8.9	6.6	11.6	8.5	2.4	13.6	7.1
Non future - uncertain (%)	2.6	0.9	2	1.8	1.9	2.7	0.8	0.5	0.3	2.1

CONCLUSION

- The annotated corpus is freely available for non commercial purposes from hlt.fbk.eu/technologies/fact-ita-bank.
- The resource has been used to develop a system based on machine learning for the identification of factuality in Italian. The tool has been evaluated on a test dataset and obtained 76.6% accuracy. This system will be integrated in the TextPro tool suite (textpro.fbk.eu/).

REFERENCES

- Tommaso Caselli, Valentina Bartalesi Lenzi, Rachele Sprugnoli, Emanuele Pianta, and Irina Prodanof. 2011. Annotating Events, Temporal Expressions and Relations in Italian: the It-TimeML Experience for the Ita-TimeBank. In *Linguistic Annotation Workshop*.
- Chantal van Son, Marieke van Erp, Antske Fokkens, and Piek Vossen. 2014. Hope and Fear: Interpreting Perspectives by Integrating Sentiment and Event Factuality. In *Proceedings of the 9th Language Resources and Evaluation Conference (LREC2014)*, Reykjavik, Iceland, May 26-31.